

AMMAN, Jordan

WMO: 402700

Lat: 31.973N

Lon: 35.992E

Elev: 779

StdP: 92.31

Time Zone: 2.00 (JOR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	3.0	-10.9	1.6	11.9	-7.9	2.1	14.4	13.1	6.8	10.8	7.5	2.6	240	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	36.1	18.3	34.8	18.2	33.2	18.0	22.1	30.5	21.1	29.5	20.3	28.8	3.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	15.8	25.6	18.6	14.8	24.7	17.6	13.9	24.1	68.7	29.9	65.1	29.3	62.1	28.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.3	7.2	-0.4	39.4	1.5	1.8	-1.5	40.7	-2.3	41.7	-3.2	42.7	-4.2	44.0		
			-2.3	23.3	1.8	1.2	-3.6	24.1	-4.6	24.8	-5.6	25.5	-6.9	26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	8.6	9.9	13.0	17.2	21.9	24.9	26.9	26.8	24.7	21.2	15.2	10.4
	DBStd	7.26	2.63	3.16	3.90	4.22	3.73	2.84	2.36	2.22	2.43	3.15	3.28	2.75
	HDD10.0	134	57	35	13	1	0	0	0	0	0	3	26	
	HDD18.3	1148	300	236	172	72	9	0	0	0	9	104	245	
	CDD10.0	3218	15	33	106	217	371	447	523	520	442	348	158	39
	CDD18.3	1189	0	1	7	37	122	197	264	262	192	99	9	0
	CDH23.3	12017	0	2	65	367	1258	2063	2902	2793	1756	751	58	1
	CDH26.7	5238	0	0	12	106	515	917	1378	1320	736	248	7	0
Wind	WSAvg	2.7	2.6	2.9	3.0	2.9	3.0	3.4	3.6	3.0	2.4	1.9	2.0	2.2
Precipitation	PrecAvg	271	63	51	49	14	3	0	0	0	8	26	53	
	PrecMax	429	235	151	156	151	23	1	0	2	35	133	173	
	PrecMin	97	2	8	0	0	0	0	0	0	0	0	1	
	PrecStd	92	43	36	35	27	5	0	0	0	10	29	50	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	23.0	28.2	32.1	36.2	37.0	38.9	38.1	36.8	33.8	27.8	22.0
		MCWB	9.8	11.5	13.9	15.2	17.1	17.8	20.0	19.4	18.3	16.7	13.6	10.2
	2%	DB	16.8	20.1	25.0	29.4	33.8	35.1	35.9	35.2	34.2	31.2	25.0	18.8
		MCWB	9.2	10.2	12.7	14.2	16.4	17.4	19.1	19.1	17.7	16.1	13.0	10.2
	5%	DB	14.9	17.4	22.2	27.4	31.8	33.1	34.0	33.9	32.2	29.4	22.9	16.8
		MCWB	8.7	9.5	11.3	13.5	15.6	17.1	18.4	19.0	17.5	15.8	12.6	9.6
	10%	DB	13.1	15.3	20.0	25.1	29.8	31.2	32.5	32.2	30.8	27.4	20.9	15.0
		MCWB	8.0	8.6	10.6	12.6	15.3	17.0	18.4	18.9	17.5	15.4	12.1	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	13.1	15.4	17.7	19.1	21.1	23.3	23.5	22.0	19.7	16.2	13.3
		MCDB	16.1	21.2	24.8	29.8	31.4	30.6	32.7	31.7	28.4	26.8	20.5	17.5
	2%	WB	10.2	11.3	13.6	15.6	17.8	20.0	21.9	22.2	20.8	18.5	15.0	11.7
		MCDB	14.5	17.6	21.9	25.6	29.6	29.9	31.2	30.4	27.9	25.6	19.6	15.7
	5%	WB	9.4	10.2	12.4	14.5	16.9	19.0	20.8	21.4	20.0	17.6	14.2	10.8
		MCDB	13.3	15.5	20.1	24.4	28.5	29.1	30.0	29.6	27.4	24.7	19.3	14.6
	10%	WB	8.7	9.3	11.3	13.4	16.0	18.2	20.0	20.5	19.2	16.9	13.3	10.1
		MCDB	12.2	14.1	18.4	23.3	27.2	28.4	29.3	28.7	26.7	23.7	18.7	13.5
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.4	9.6	11.3	11.9	11.7	11.1	11.4	11.5	10.8	9.5	8.0
		MCDBR	10.7	12.3	13.1	13.9	14.2	13.4	12.4	12.4	13.3	13.1	12.0	10.9
		MCWBR	5.6	6.0	6.2	6.1	5.9	6.0	6.2	6.1	5.5	5.3	5.3	5.7
	5% WB	MCDBR	9.0	10.2	11.5	12.7	12.5	12.0	11.5	11.8	11.6	11.1	9.5	8.5
		MCWBR	5.3	5.5	6.2	6.3	6.1	6.4	6.4	6.4	5.7	5.3	5.1	5.0
Clear-Sky Solar Irradiance	taub	0.344	0.373	0.417	0.461	0.460	0.389	0.396	0.403	0.397	0.416	0.366	0.346	
	taud	2.265	2.168	2.027	1.913	1.943	2.178	2.167	2.150	2.168	2.100	2.225	2.270	
	Ebn at Noon	865	873	858	834	836	894	885	876	867	814	830	839	
	Edh at Noon	109	133	165	193	189	149	150	150	142	141	113	104	
All-Sky Solar Radiation	RadAvg	3.08	3.89	5.29	6.44	7.49	8.32	8.06	7.39	6.32	4.81	3.60	2.89	
	RadStd	0.19	0.30	0.30	0.30	0.31	0.12	0.09	0.10	0.16	0.19	0.18	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (17 neighbors)	+0.48	N/A	-2.69	N/A	-0.98	-1.38	N/A	-85	+157	+101
		+0.44	N/A	-1.95	N/A	N/A	N/A	N/A	-60	+140	+92

Nomenclature: See separate page